ahmad_mahmoud.resume.ts

CTO · HEAD OF ENGINEERING ● 12+ YRS

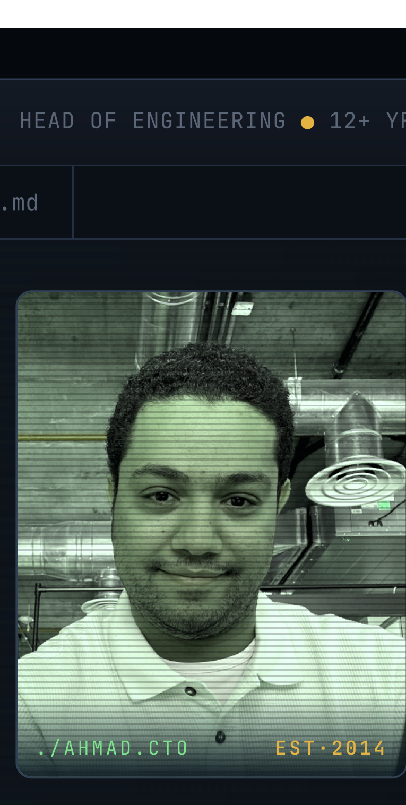
● cto.ts | experience.json | projects/ | stack.config | README.md

→ whoami --full

Ahmad Abdul-Jawad Mahmoud

Chief Technology Officer / Head of Engineering

// **12+ years** building · **8+ leading teams** · Cairo → Riyadh · millions of users across MENA



12+	8+	35M+	20k+
YEARS ENGINEERING	YEARS LEADING TEAMS	USERS REACHED	BUSINESSES ON THE POS I LED

```
/**
* Chief Technology Officer — 12+ years building, 8+ leading.
* Seven of them running the iOS org behind a POS platform trusted by 20k+
* restaurants and cafés: technology strategy, architecture and roadmap, the
* standards a team runs on, and the answer for delivery.
* I have also carried the whole technical scope of a founding CTO alone —
* a production ride-hailing platform shipped solo across iOS & Android,
* a NestJS / PostgreSQL / Redis backend and a Next.js ops console — so no
* layer of the stack is a black box when I am deciding for a company.
*/
```

## technology_office()

4 pillars

[01] technology strategy

Choose the bets, own the consequences

Stack, architecture and roadmap as one decision — platform choices, module boundaries, data models, security posture, and **migrations that keep a codebase current without stalling the business.**

◆ 7-year production POS moved to SwiftUI mid-flight

[02] organisation & talent

An engineering function, not a headcount

Seven years building and leading a **5+ engineer team**: onboarding, mentoring, career growth, and a review culture where quality is the default.

◆ 99.8–99.9% crash-free under enterprise load

[03] delivery & operations

Commitments the company can plan around

CI/CD and release automation built from nothing, suites that gate merges, explicit performance and crash budgets, planning the team can keep.

◆ 60% faster deployments · 40% faster load times

[04] whole-stack ownership

No layer is someone else’s problem

iOS & Android clients, NestJS/PostgreSQL/Redis backends, Next.js ops consoles — **all four layers of a production platform built solo.**

◆ 4 platforms · 1 contract · 1 engineer

## experience()

8 roles

iOS Team Lead @ Marn — Jahez Group

Aug 2019 → Aug 2026

RIYADH, KSA

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the review culture behind an enterprise POS used by **20k+ restaurants** and cafés.
- **Owned the technical roadmap and architecture:** an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Set the engineering standards** — module boundaries, code review, automated testing — holding the app at a **99.8–99.9% crash-free** rate in production.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release process from scratch**, cutting deployment time **60%** and shortening every engineer's feedback loop.
- Ran the performance programme (**40% faster load times**) and designed the table-management, order-processing and inventory systems at the product's core.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments.

Founding Platform Architect — iOS · Android · Backend · Web @ Ride-Hailing Platform · Independent

Mar 2026 → Present CURRENT

Took the **full technical scope of a founding CTO** and carried it alone — shipped a production ride-hailing platform **solo**: two native **iOS/SwiftUI** apps (Rider & Driver) plus their Android counterparts, a hardened backend, and a real-time ops console — **owning every layer** and setting the engineering standards (idempotency, double-entry accounting, acyclic module boundaries, CI test suite).

› **Backend APIs**
// NestJS · TypeScript · Mar '26 → now

Modular monolith, ~17 domains — **trip state machine**, **scored dispatch** over a Redis geo index, **double-entry ledger**, realtime rider/driver/admin **sockets**, JWT + phone-OTP, idempotency & audit log · **1,150+ tests**, Swagger docs.

› **Admin Console**
// Next.js 14 · Jun '26 → now

Bespoke **19-screen** ops surface — live **MapLibre** trips map, dispatch/SOS live boards, KYC review, finance reconciliation, refunds/payouts, map **zone editor**, **Google SSO**.

› **iOS Apps · Rider & Driver**
// SwiftUI · Jul '26 → now

Two native **SwiftUI** apps — live GPS **trip lifecycle**, realtime sockets, in-app chat, payments, and driver earnings & turn-by-turn navigation.

› **Android Apps · Rider & Driver**
// Kotlin · Jetpack Compose · Sep '26 → now

Native **Kotlin / Compose** parity for both apps — same trip lifecycle, realtime tracking, chat, and payments across **one shared API contract**.

One backend · one contract · four clients — plus growth loops (promotions, referrals, loyalty, wallet), driver safety/SOS, and scheduled reconciliation jobs, shipped end-to-end.

Senior iOS Engineer @ HaramBlur · voluntary charity project

Dec 2025 → Jun 2026

- Designed an architecture integrating **on-device machine learning** for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

iOS Team Lead @ Ego — Ride Hailing

Mar 2018 → May 2019

RIYADH, KSA

- Developed a ride-hailing ecosystem (iOS, watchOS) serving **15M+ users** in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Senior iOS Engineer @ HungerStation — DeliveryHero

Sep 2016 → Mar 2018

KSA

- Contributed to a major food-delivery platform serving **20M+ users**.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

iOS Developer @ ALNapedh

May 2016 → Aug 2016

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

iOS & Apple Watch Developer @ Nile Design Co.

Jul 2015 → Sep 2015

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

iOS Developer @ UK Pro. Solutions Ltd.

May 2014 → May 2015

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

## projects[]

selected

[01] ai / privacy

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur, live video monitoring.

◆ AI-powered privacy browser

[02] full-stack

Ride-Hailing Platform

SwiftUI · Kotlin · Jetpack Compose · NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js

End-to-end ride-hailing system built solo: **four native apps** (Rider & Driver on iOS + Android), a hardened NestJS backend (idempotency, double-entry ledger, 1,150+ tests), and a real-time Next.js ops console.

◆ Full-stack ownership · prod security

[03] delivery

HungerStation

UIKit · Firebase · Real-time APIs · Perf

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

[04] mobility

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · watchOS

Complete ride-hailing ecosystem: rider, driver, and Apple Watch apps. Real-time GPS and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

[05] enterprise

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 20k+ restaurants across Saudi Arabia. Offline-first architecture, complex table management, real-time orders.

★ 4.8 / 5 — 2,437 ratings

[06] community

Daleelo

MapKit · Core Location · Social SDKs

Muslim community app — prayer times, mosque finder, and community features with enhanced location + social integration.

★ 4.5 / 5 — 1,287 ratings

## stack.config

leadership =
Team Leadership · Technical Direction · Architecture Governance · Tech Roadmap · Mentoring & Career Growth · Onboarding · Code Review Culture · Estimation & Planning · Cross-functional · Quality & Crash Budgets · Release Management · Agile/Scrum

ios_2025 =
Swift 6.0 · SwiftUI · UIKit · Combine · async/await · SwiftData · iOS 18+

android_2025 =
Kotlin · Jetpack Compose · Android · Coroutines · Flow · MVVM

architecture =
MVVM · VIPER · Clean Architecture · Coordinator · TDD

apple_ecosystem =
watchOS · tvOS · macOS · WidgetKit · Live Activities · App Intents

intelligence =
Core ML · Vision · ARKit · Dynamic Island · App Tracking

backend_apis =
NestJS · TypeScript · PostgreSQL · TypeORM · Redis · JWT/OTP · REST · GraphQL · Firebase · CloudKit · WebSockets · Swagger

tooling =
Xcode 15+ · Fastlane · TestFlight · Instruments · Git · Copilot · Cursor

## portfolio/

FOOD & GAMING
- HungerStation
- Punters

TRANSPORTATION
- Ego — Rider
- Ego — Driver
- Ego — Watch

ENTERPRISE
- Marn POS
- Medical Guide

SOCIAL & COMMUNITY
- Daleelo
- NU-inc
- VeeU

WATCHOS
- Ego Watch
- TaQs Weather
- TweetMe

E-COMMERCE
- iBss
- Exclusive Luxe
- Scratch

RELIGIOUS & EDU
- Afasy
- ArabNileGroup

UTILITY & TOOLS
- Unlocking Smart
- Insta
- iContacts
- First iOS app

AI & ML
- HaramBlur
- RecognizeMe
- Grad project · OpenCV

WEB
- FaceDemo
- Web Pages

## record{}

// education

B.Sc. Information Systems
Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

// certification

ISTQB Foundation Level
Egyptian Software Testing Qualification Board

// languages

Arabic	native
English	professional

## honours[]

01 Led an iOS engineering team for **7 consecutive years** on a POS platform serving **20k+ restaurants** and cafés.

02 Held a **99.8–99.9% crash-free** rate and cut deployment time **60%** with the standards, tests and CI/CD the team ran on.

03 Architected a **full-stack ride-hailing platform solo** — four native apps (iOS + Android), NestJS/PostgreSQL backend, and a Next.js ops console.

04 Built a food-delivery platform serving **20M+ users** and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) & early SwiftUI adopter (WWDC 2019).

main | ● CTO · Head of Engineering | Cairo → Riyadh | UTF-8 | TypeScript · Swift · Kotlin | ★ 12+ yrs · 8+ leading